Phase 1 Implementation Audit
Revision 44 Blueprint vs. Jupyter Notebook

Compliance Verification Report

May 3, 2026

Overall Verdict

✗ Document spec is 100% compliant, but the ACTUAL code in the notebook is ∼0% compliant with Phase 1.

The `.tex`/`.pdf` blueprint (revision 44) now contains all 5 required fixes, but the attached Jupyter notebook still runs the **revision 42 configuration verbatim** — none of the three Phase 1 P0 actions have been switched on in code.

1 Critical Findings

Finding 1 ✗ — τ is still the old learnable value, NOT static $\tau = 0.3$

The notebook passes `temperature = 0.1` and the comment explicitly says "initialisation of the learnable tau (Revision 9 spec)". There is no `-temperature 0.3` flag and no `-temperature-mode static` flag. The training log's `tau_final` is still being parsed as a drifting value, confirming it is still learnable.

```
temperature: float = 0.1  # initialisation of the learnable tau
                          # (Revision 9 spec, Section 3.1)
```

```
temperature: float = 0.3                # Phase 1: static anchor
learnable_temperature: int = 0          # NEW FLAG -- force static
```

On the CLI side, add `-learnable-temperature 0` (or whatever flag `train.py` exposes). If `train.py` does not expose such a flag yet, the repository's `model.py` must be patched so `self.tau` becomes either `nn.Parameter(torch.tensor(0.3), requires_grad=False)` or a plain buffer.

Finding 2 ✗ — AVRF is STILL ON (`damps_avrf = 1`)

Required change: The notebook sets `dampsavrf: int = 1 # Logit-clipped Wiener gate`. Phase 1 action #3 explicitly requires AVRF **disabled** (`-damps-avrf 0`). This is the single most consequential miss — the entire rationale for the Phase 1 combined +5%~8% gain depends on this flag being flipped.

```
dampsavrf: int = 0     # Phase 1: AVRF ablation OFF
```

Finding 3 ! — Eval protocol audit is NOT documented in the notebook

Required change: The spec demands a 7-point audit: (i) 5-core, (ii) split ratio, (iii) all-ranking vs. sampled, (iv) NDCG $\log_2$ convention, (v) ID remapping, (vi) popularity filtering, (vii) sort stability. The notebook contains **zero audit evidence** — no assertions, no comparison prints against MMHCL reference values, and no printed confirmation that each of the 7 items was checked.

```python
# === Phase 1 Item #1: Eval Protocol Audit ===
from damps.data_utils import load_clothing_split
split = load_clothing_split(DATADIR)
assert split['core_filter']   == 5,                   "FAIL (i) 5-core"
assert split['train_ratio']   == 0.8,                 "FAIL (ii) split ratio"
assert split['split_seed']    == 2023,                "FAIL (ii) split seed matches MMHCL"
assert split['eval_mode']     == 'all_ranking',       "FAIL (iii) must be all-ranking"
assert split['ndcg_log_base'] == 'log2_i_plus_2',     "FAIL (iv) NDCG convention"
assert split['id_remap_consistent'],                  "FAIL (v) ID remap"
assert split['mask_train_at_test'],                   "FAIL (vi) popularity filter"
assert split['stable_sort'],                          "FAIL (vii) tie stability"
print("OK  All 7 eval-protocol checks match MMHCL reference.")
```

Finding 4 ✗ — Combined fix variant not scheduled

Required addition — insert a new cell before training: The stop-gate in Section 5.2 of revision 44 requires **4 variants**: (a) anchor, (b) $+\tau = 0.3$, (c) + AVRF off, (d) combined. The notebook only runs a single configuration $\times$ 10 seeds. It is missing the loop over 4 configurations.

Required change: Wrap the seed loop in an outer configuration loop:

```python
phase1_variants = {
    'anchor'   : {'temperature': 0.1, 'dampsavrf': 1, 'learnable_temp': 1},
    'tau03'    : {'temperature': 0.3, 'dampsavrf': 1, 'learnable_temp': 0},
    'avrf_off' : {'temperature': 0.1, 'dampsavrf': 0, 'learnable_temp': 1},
    'combined' : {'temperature': 0.3, 'dampsavrf': 0, 'learnable_temp': 0},  # recommended
}
for variant_name, overrides in phase1_variants.items():
    for seed in seeds:   # 10 seeds each
        ...
```

Finding 5 ✗ — Paired t-test + Bonferroni script not invoked

Section 5.2 mandates `scripts/paired_t_test.py` with Bonferroni correction when more than one variant is compared against the anchor. The notebook's "Results Summary" cell only reports mean $\pm$ std; it never runs a paired t-test against the anchor. The header text "must average across 10 seeds with 95% CIs and paired t-tests vs. the MMHCL baseline (`scripts/paired_t_test.py`)" confirms the script exists in the repository but the notebook does not call it.

```python
subprocess.run([PYTHON_EXE, "scripts/paired_t_test.py",
                "--anchor",   "../dataset/anchor_summary.json",
                "--variants", "../dataset/tau03_summary.json",
                              "../dataset/avrf_off_summary.json",
                              "../dataset/combined_summary.json",
```

```
                "--bonferroni", "1",
                "--alpha",      "0.05"], check=True)
```

Finding 6 ! — Notebook header still references "Revision 9 / revision42.tex"
Required addition — new cell after results aggregation: The notebook's Markdown header cell says "follows the Revision 9 (100% Compliance Check — Final Lock) architecture spec (see `DAMPS_to_MMHCL_architecture_revision42.tex/.pdf`)". It should now reference revision 44. This is cosmetic, but will confuse any reviewer who opens the notebook expecting revision 44 logic.

2 Compliance Scorecard

Phase 1 Requirement	**Document (rev44)**	**Notebook / Code**	**Status**
Scope restricted to Phase 1	✓	✓	Pass
Action #1 — 7-point eval audit executed	✓ specified	✗ no audit cell	**FAIL**
Action #2 — Static $\tau = 0.3$	✓ specified	✗ `temperature`=0.1, still learnable	**FAIL**
Action #3 — AVRF disabled	✓ specified	✗ `dampsavrf`=1	**FAIL**
Quantitative stop-gate (0.0870 / 0.0900)	✓	✗ not checked in code	**FAIL**
4 variants × 10 seeds	✓	✗ 1 variant × 10 seeds	**FAIL**
Paired t-test + Bonferroni	✓	✗ not invoked	**FAIL**
Anchor numbers (0.0786 / 0.0383)	✓	! only `recall20`=0.0881 paper value hard-coded	Partial
Notebook header references rev44	n/a	✗ says rev42 / Revision 9	Cosmetic

3 Minimum Patch to Reach 100% Compliance

Single most important change. In the notebook's training-config cell, three lines suffice to flip both experimental interventions:

```
temperature:    float = 0.3   # was 0.1 (Phase 1 static tau)
dampsavrf:      int   = 0     # was 1   (Phase 1 AVRF ablation)
# plus a new CLI arg to freeze tau, e.g.:
learnable_temp: int   = 0     # new; forces static tau in train.py
```

Then add the three new cells above:

1. Eval-protocol audit assertions (before training).
2. Outer configuration loop over the 4 Phase 1 variants.
3. `scripts/paired_t_test.py` invocation with Bonferroni after aggregation.

Why this matters. Without these edits, running the notebook as-is will simply reproduce the revision-42 mean of 0.0786 ± 0.0016 a second time — Phase 1 will yield **zero** of its projected +5%~8% gain because *neither* of its two experimental interventions ($\tau = 0.3$, AVRF off) is actually active in the code path.

4 Conclusion

The blueprint (revision 44) is compliant; the implementation (notebook) is not. Please apply the six edits above before launching the 10-seed Phase 1 campaign — otherwise the runs will consume GPU hours without testing any of the Phase 1 hypotheses.